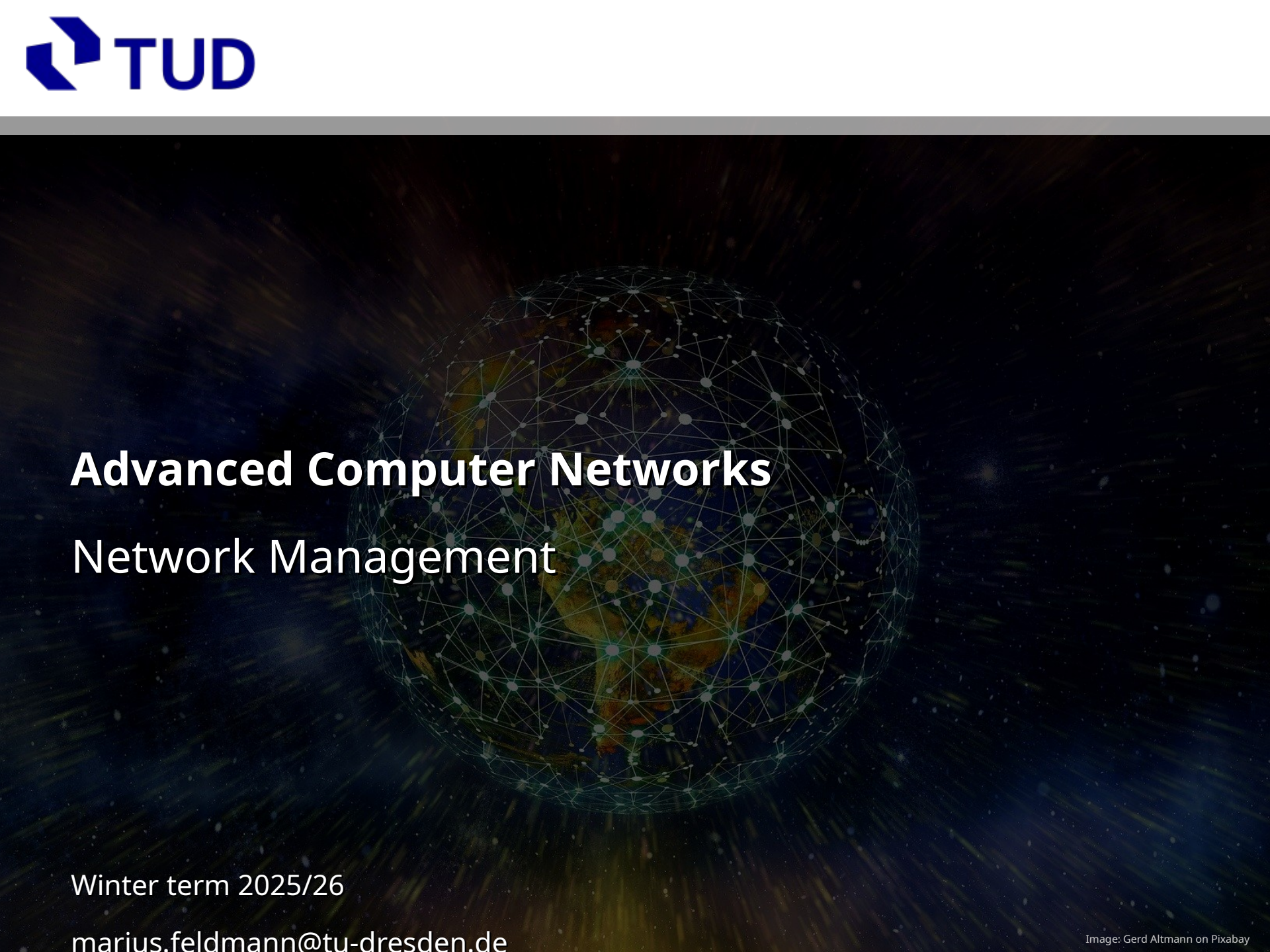


The background of the slide is a dark, starry space with a central globe. The globe is covered in a complex network of white lines and dots, representing a global computer network. The text "Advanced Computer Networks" and "Network Management" is overlaid on the left side of the globe.

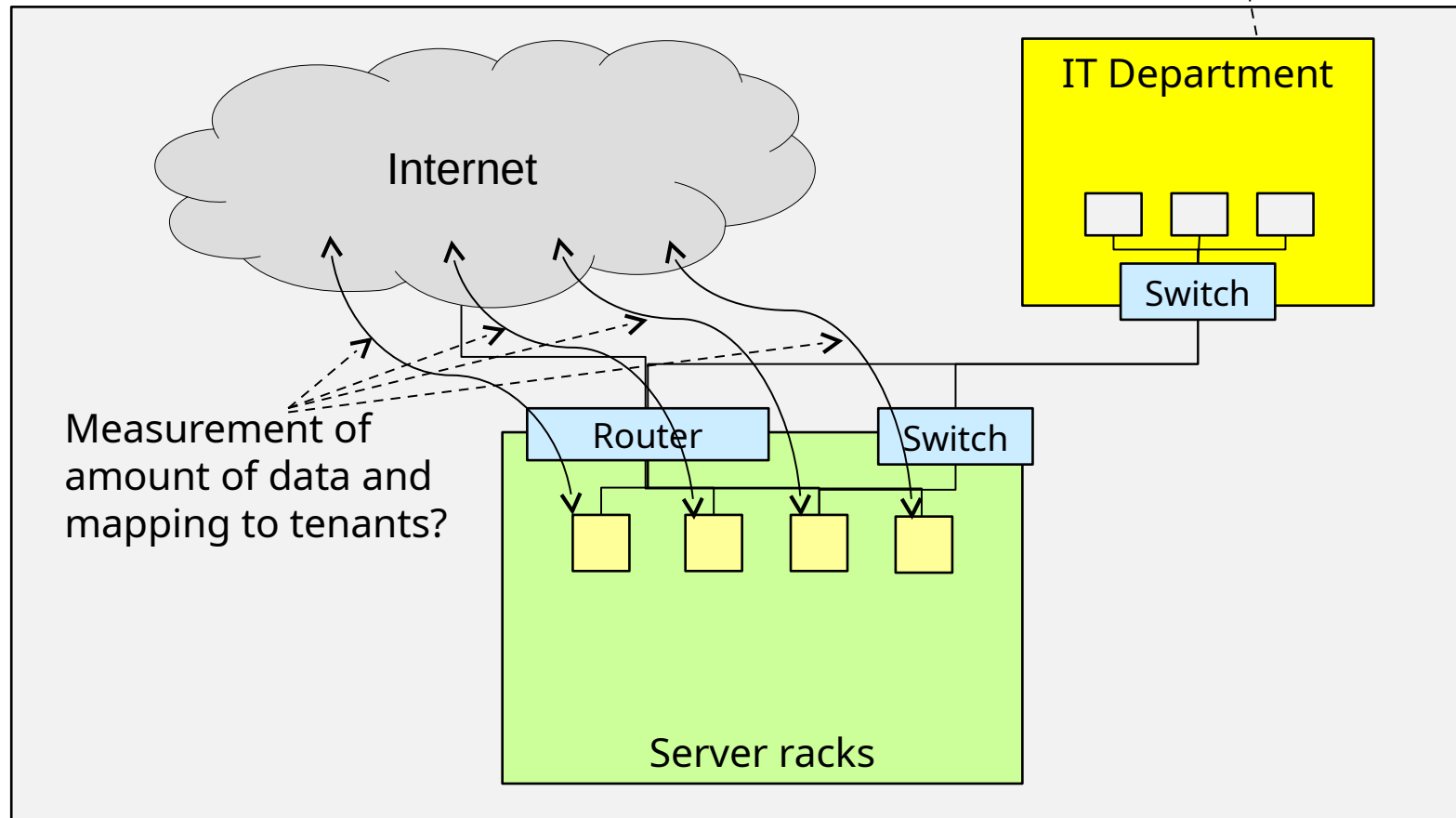
Advanced Computer Networks

Network Management

Winter term 2025/26

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Introduction



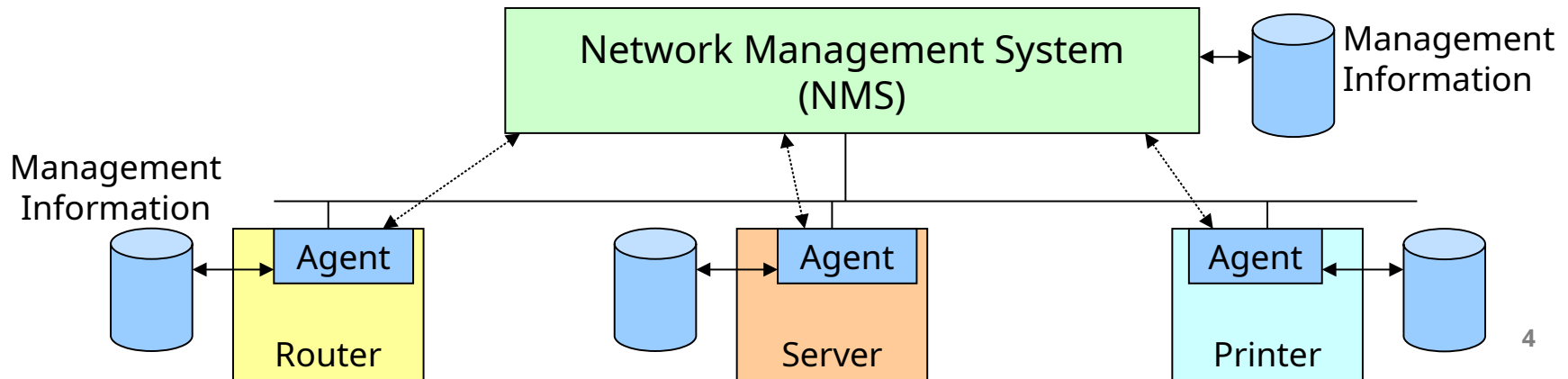
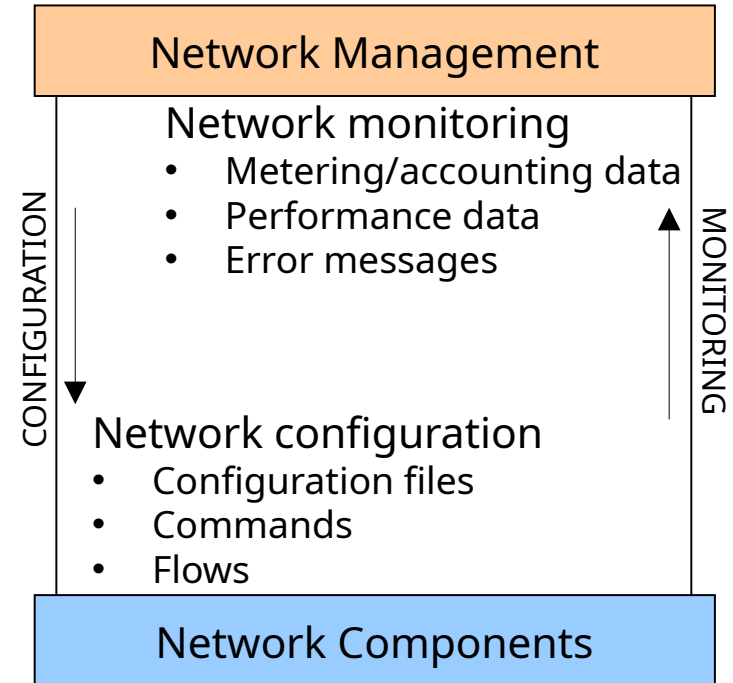
- Which tasks does network management cover?
- Characteristics of a network management solution?
- Protocols and approaches for network management?

Outline

1. Overview
2. FCAPS model
3. Out-of-band- / in-band network management
4. Characteristics / functions of network management solutions
5. Simple Network Management Protocol
 - Overview
 - Management Information Base
 - Structure of Management Information
 - Protocol
 - SNMPv3
6. Remote Network Monitoring
7. sFlow / NetFlow / IPFIX
 - IPFIX protocol details
 - IPFIX example
8. NETCONF
9. Monitoring based on Prometheus

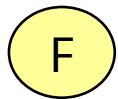
Network Management

- Informal: Network management refers to the purposeful usage of activities, methods, processes and tools to maintain networks and to continuously guarantee the networks' availability based on given requirements (SLAs)
- Network management covers
 - Passive, observation tasks
 - Active, modifying tasks
- Network management =
Configuration + Monitoring

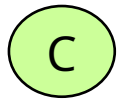


FCAPS model

- FCAPS model is a model specified by the International Organization for Standardization and describes the categories of tasks covered by network management



Fault Management: Detection, isolation, diagnosis and correction of failures in the network



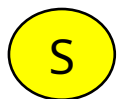
Configuration Management: Discovery of the systems available in the network, controlling these systems, collecting all relevant configuration data/files and provisioning of configuration data/files



Accounting Management: Aggregation of metering information and collection of this information with the purpose of a tenant-specific accounting



Performance Management: Measuring and log end-to-end performance and performance of the various network components, specification and execution of performance benchmarks and specification of threshold values and their monitoring

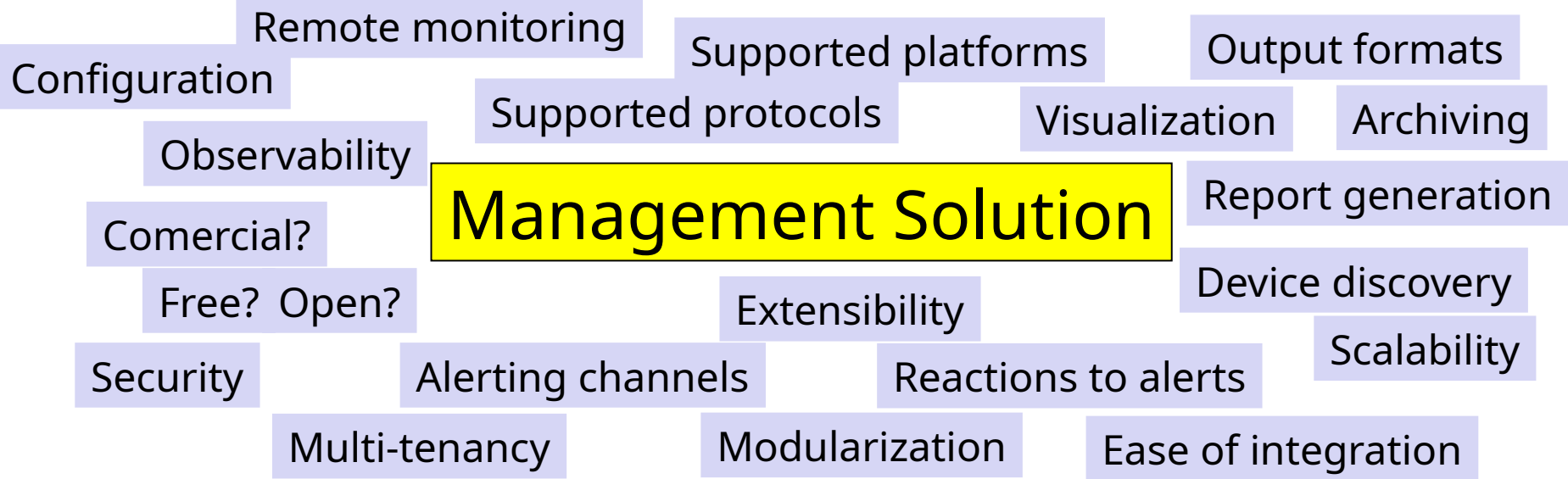


Security Management: Definition of security targets, availability monitoring, creation and management of security measures

Out-of-band / In-band Network Management

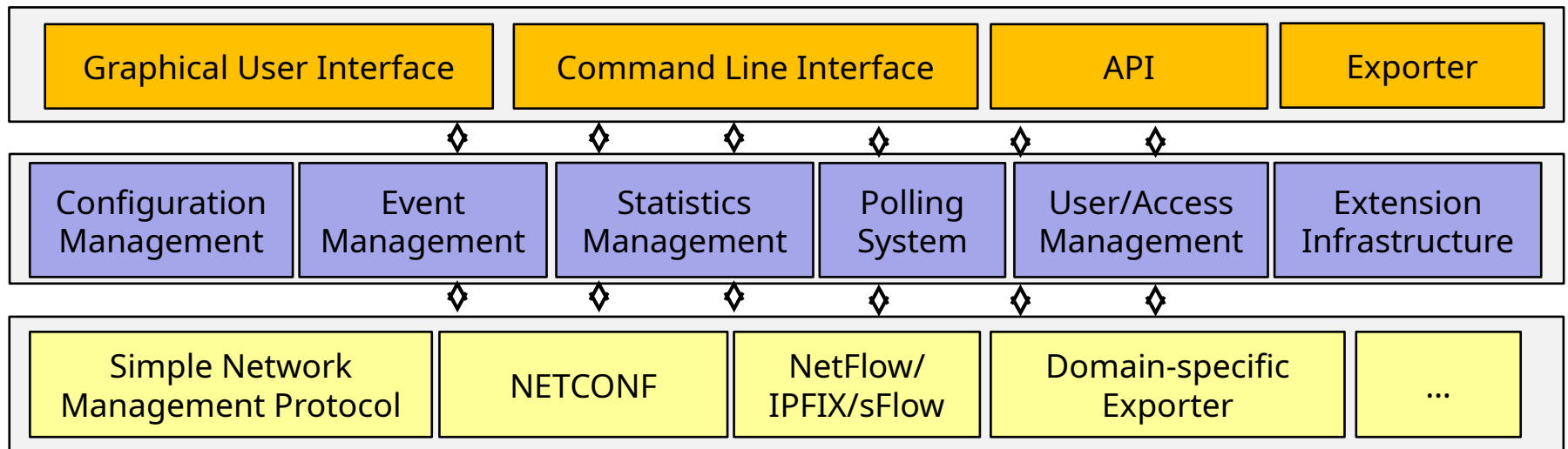
- **Out-of-band network management:**
 - Strict separation of productive network and management network infrastructure (→ separate network for fulfilling the management tasks)
 - Example: Non-IP-based management network
 - Advantage: Creating a bridge between networks is more difficult; separation of load in the networks
 - Disadvantage: Additional knowledge required
 - Example: IP-based management network
 - Advantage: Existing experiences can be used for the management network; separation of load in the networks
 - Disadvantage: Creating a bridge between the networks is simpler
-
- **In-band network management**
 - Productive network and network for monitoring and configuration are implemented on top of the same infrastructure
 - Advantage: Existing infrastructure is reused
 - Reduced installation and configuration effort
 - Cost reduction
 - Disadvantage: Failures in the productive network may result in negative impact on the management network

Characteristics and functions of a NMS



3-Tier Architecture:

Interfaces

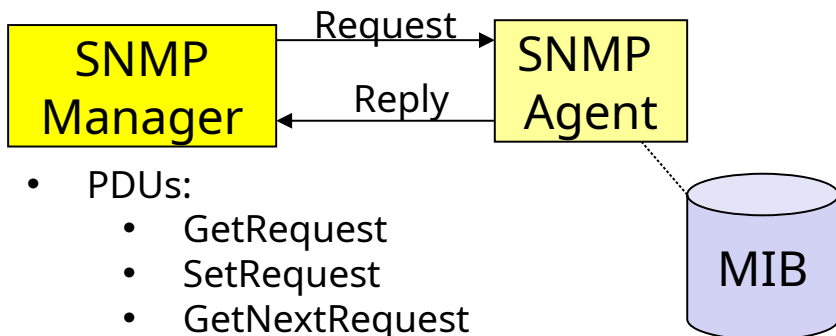


Data Sources

Simple Network Management Protocol (SNMP) - Overview

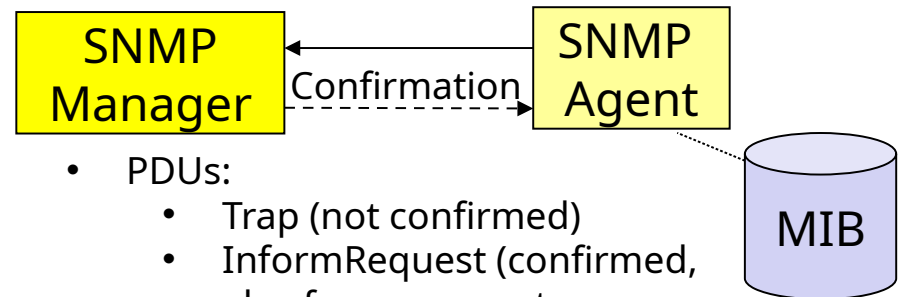
- Simple Network Management Protocol framework allows management of devices in IP networks
- Three versions have evolved
 - SNMPv1, SNMPv2c (c=community-based), SNMPv3
 - SNMPv1 and SNMPv2c are still applied, however, they have been declared as “historic”
- Framework specification covers four foundations:
 - Specification language for data (Structure of Management Information (SMI))
 - Management information (Management Information Base (MIB))
 - Protocol operations
 - Security and administration mechanisms
- Protocol supports two communication patterns:

1. Synchronous request/reply communication



- PDUs:
 - GetRequest
 - SetRequest
 - GetNextRequest
 - GetBulkRequest
 - (+Response)

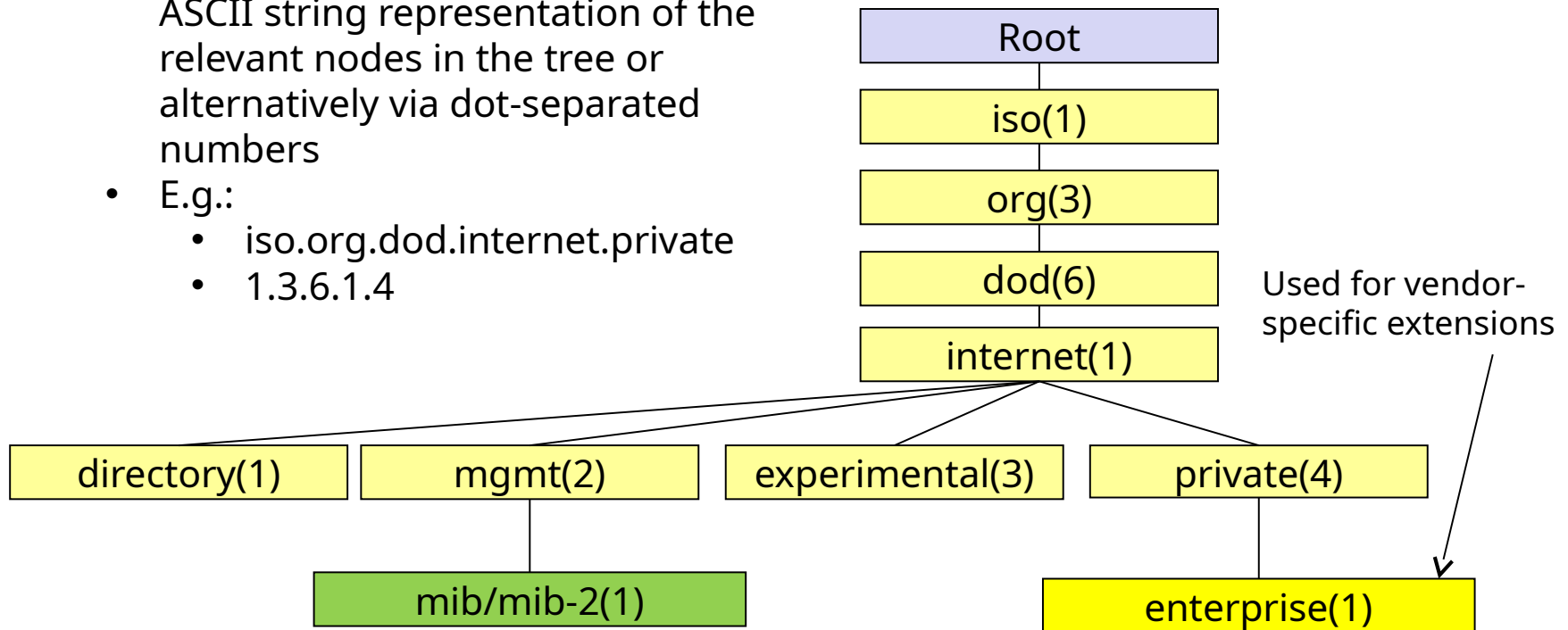
2. Asynchronous communication



- PDUs:
 - Trap (not confirmed)
 - InformRequest (confirmed, also for manager-to-manager communication)

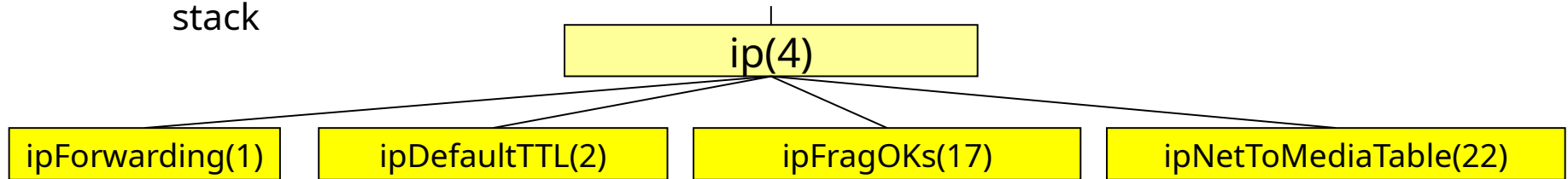
Management Information Base (MIB)

- Managed Objects are represented in a MIB tree and are made accessible via this tree structure
- RFC 1155 defines overall structure of the MIB for typical networked systems
- MIB II is defined as "Management Information Base for Network Management of TCP/IP-based internets: MIB-II" in RFC 1213
- Objects are accessible via Object Identifier (OID)
- OID is specified via dot-separated ASCII string representation of the relevant nodes in the tree or alternatively via dot-separated numbers
- E.g.:
 - iso.org.dod.internet.private
 - 1.3.6.1.4

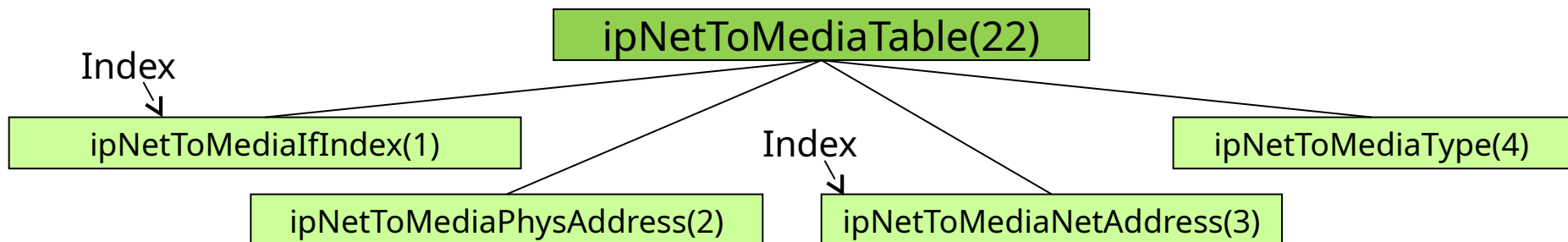


MIB II – selected details

- Example for the level of details that is available via SNMP about the networking stack



- ipForwarding: Specifies, if the system is a router
- ipDefaultTTL: Default TTL value
- ipFragOKs: Number of successfully fragmented IP packets
- ipNetToMediaTable: Table with mapping between IP and MAC addresses



- ipNetToMediaIfIndex: Interface index
- ipNetToMediaPhysAddress: Physical (MAC) address associated to the IP address of the entry
- ipNetToMediaNetAddress: IP address of the entry
- ipNetToMediaType: Type of IP address assignment (invalid, dynamic, static, other)

Management Information Base - Tables

- One or more columns may be declared as index of the table
- OID: Base_OID.table.1.selectedColumn.Index
- The 4th hierarchical level is specified using the content of all index columns of the relevant row
- Example:
 - Usage of an IP address as index – in the 4th hierarchical level the address including dots is used

Table(4)				
Column definition (incl. type information)(1)				
	(1)	(2)	(3)	(4)
(1)	Cell(1)	Cell(1)	Cell(1)	Cell(1)
(2)	Cell(2)	Cell(2)	Cell(2)	Cell(2)
(3)	Cell(3)	Cell(3)	Cell(3)	Cell(3)

Cell(3)	Base_OID.4.1.1.3	Cell(2)	Base_OID.4.1.3.2
---------	------------------	---------	------------------

Structure of Management Information Version 2 (SMIV2)

- MIB is specified using the Structure of Management Information Version 2 (defined in RFCs 2578, 2579, 2580)
- SMI is formed by a subset of the *Abstract Syntax Notation One* (ASN.1), a standard for representing encodings of data in the telecommunication and networking domain
- SMI makes it possible to define and group objects, data types and meta data about these objects
- Example taken from RFC 1213, page 38 f.

ipNetToMediaEntry OBJECT-TYPE

SYNTAX IpNetToMediaEntry

ACCESS not-accessible

STATUS mandatory

DESCRIPTION

"Each entry contains one IpAddress to 'physical' address equivalence."

INDEX { ipNetToMediaIfIndex,
ipNetToMediaNetAddress }

::= { ipNetToMediaTable 1 }

↖
Type declaration of table (via
structure "IpNetToMediaEntry")

SMI - Examples

```
IpNetToMediaEntry ::=
    SEQUENCE {
        ipNetToMediaIfIndex
            INTEGER,
        ipNetToMediaPhysAddress
            PhysAddress,
        ipNetToMediaNetAddress
            IpAddress,
        ipNetToMediaType
            INTEGER
    }
```

Defines element types of the four table columns

```
ipNetToMediaNetAddress OBJECT-TYPE
    SYNTAX IpAddress
    ACCESS read-write
    STATUS mandatory
    DESCRIPTION
        "The IpAddress corresponding to the
        media-dependent 'physical' address."
    ::= { ipNetToMediaEntry 3 }
```

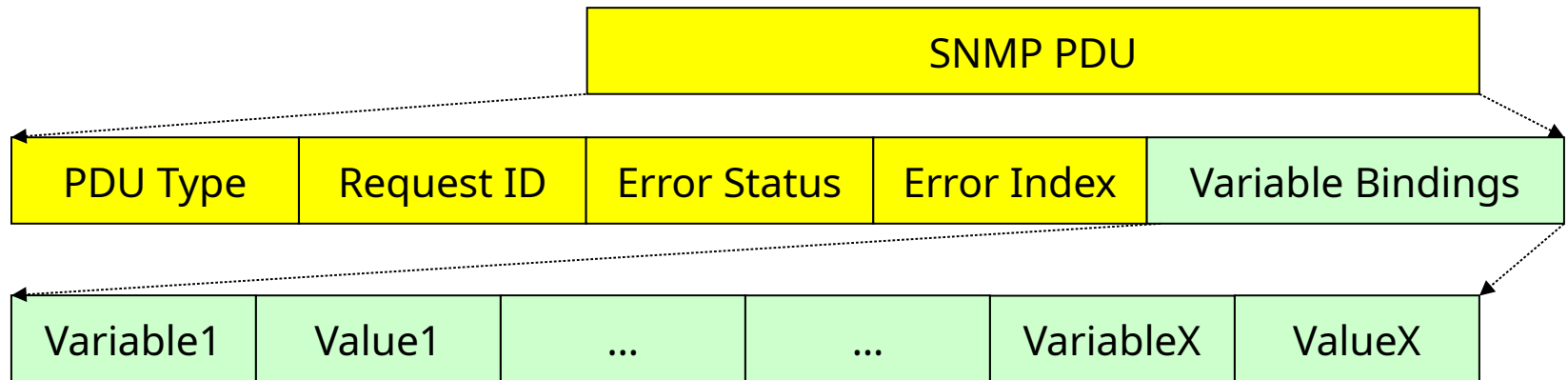
Specifies access rights ("read-write") and if the object has to be provided in der MIB ("mandatory")

-
- Example for a trap definition (from RFC 3418, p. 10)

```
coldStart NOTIFICATION-TYPE
    STATUS current
    DESCRIPTION
        "A coldStart trap signifies that the SNMPv2 entity, acting in an agent role, is
        reinitializing itself and that its configuration may have been altered."
    ::= { snmpTraps 1 }
```

Simple Network Management Protocol (SNMP)

- Typical transport protocol: UDP
- Seven types of PDUs are distinguished (see slide 8)



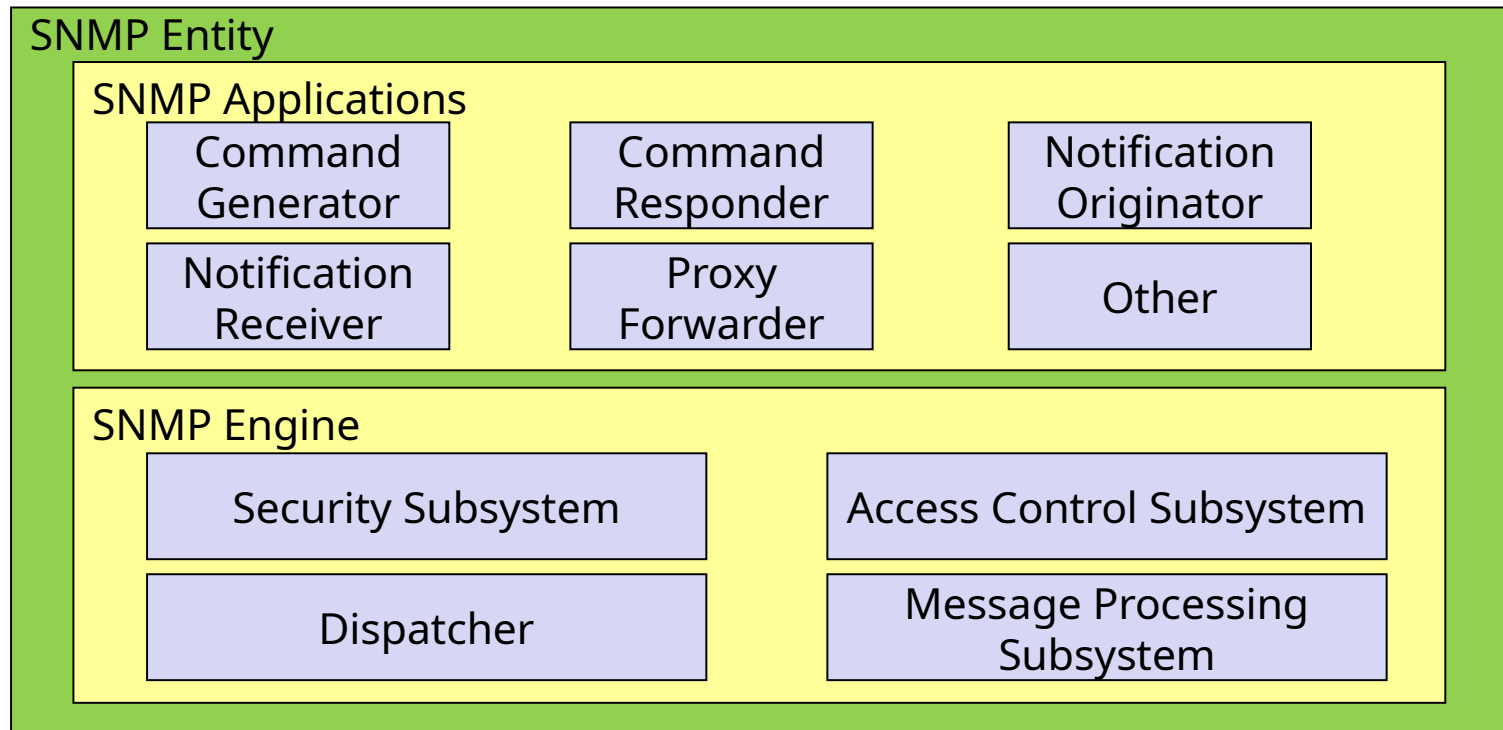
```
get(1.3.6.1.2.1.1.5.0)
response(error-status=noError; 1.3.6.1.2.1.1.5.0 => hostname)

set(1.3.6.1.2.1.1.5.0 => newHostname)
response(error-status=noError; 1.3.6.1.2.1.1.5.0 => newHostname)

set(1.3.6.1.2.1.1.5.0 => anotherHostname, 1.3.6.1.2.1.1.5.2.3 => 0)
response(error-status = noSuchName; error-index=2)
```

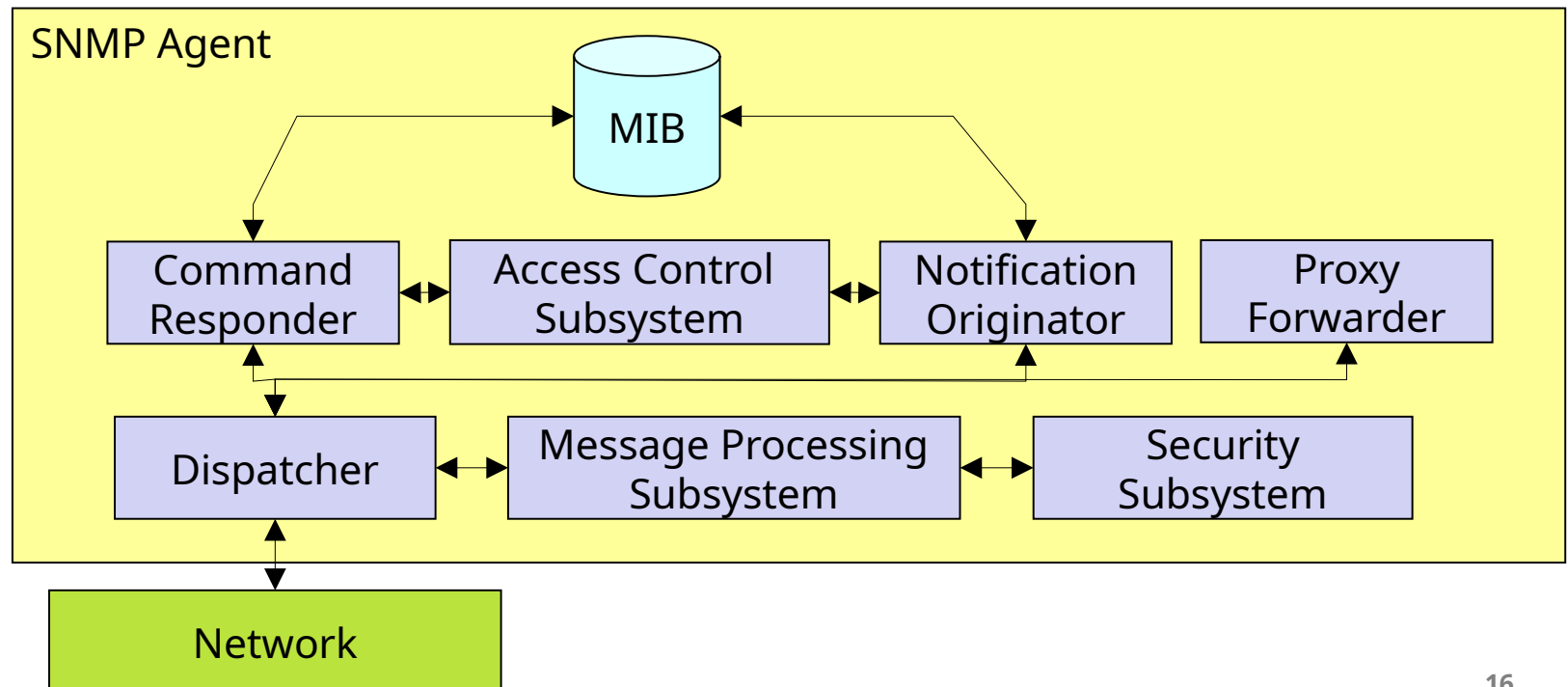
SNMPv3 architecture

- SNMPv3 architecture (chapter 3.1. in RFC 3411) is highly modular and flexible
- Security aspects have been addressed in contrast to former versions
- Components can be classified into two categories:
 - Application Entities
 - Protocol Entities (part of the SNMP engine)



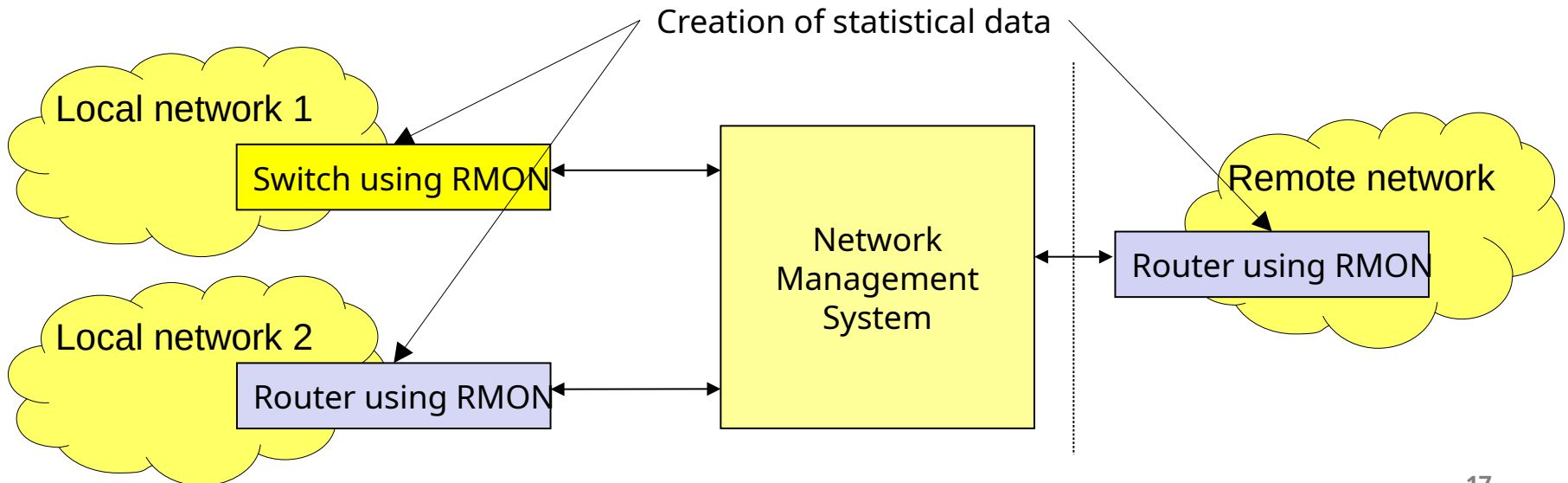
SNMPv3 architecture

- Classical SNMP managers and agents can be build based on the overall architecture
- SNMP Manager: SNMP entity, which contains one or more command generator and notification receiver applications
- SNMP Agent: SNMP entity, which contains one or more command responder and notification originator applications
- Example: SNMP agent formed based on the SNMPv3 architecture (see RFC 3411, p. 23)



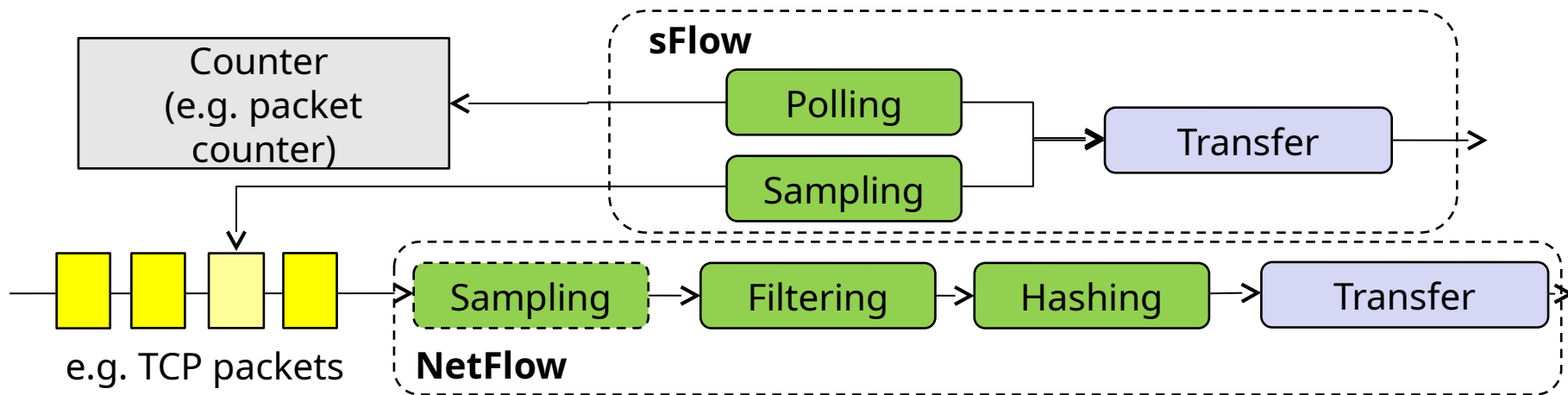
Remote Network MONitoring (RMON)

- Monitoring and protocol analysis standard based on SNMP and the MIB II
- Used to aggregate statistical data and forward them to a manager
- Overview provided via RFC 3577
- Statistical data may be created for a single device as well as for the overall network
- Goals:
 - Provide options for offline operations
 - Reduce overhead of management data
 - Foster hierarchical network management
 - Proactive monitoring / management and early error detection

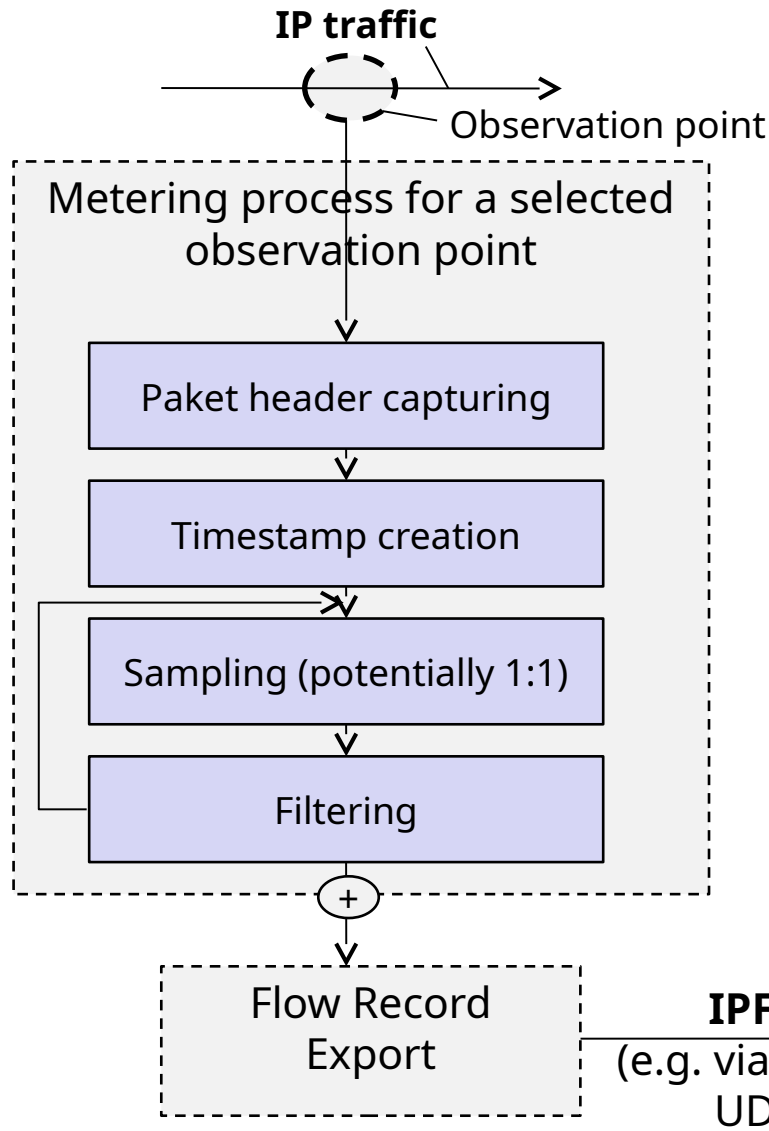


sFlow / NetFlow / IPFIX

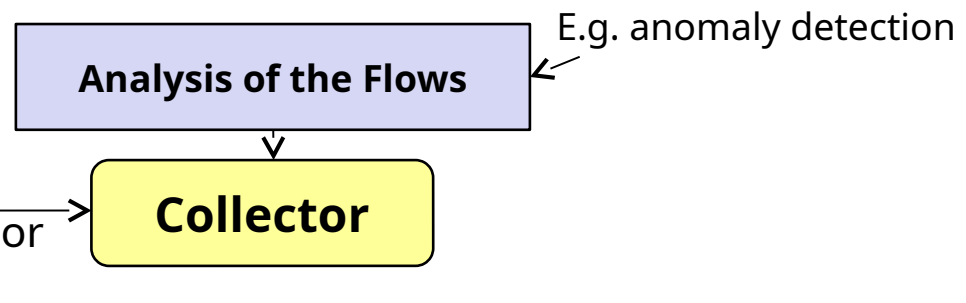
- SNMP is not applicable for continuous transfer of IP network traffic to a central instance
- E.g. goals of such data transfer:
 - Accounting
 - Evaluation of data to detect potential attacks (Intrusion Detection System)
- Relevant approaches:
 - sFlow: highly scalable approach for transferring counters and additional information about data flows based on sampling
 - NetFlow/IPFIX: Approach to transfer information about data flows (packets) with optional sampling support



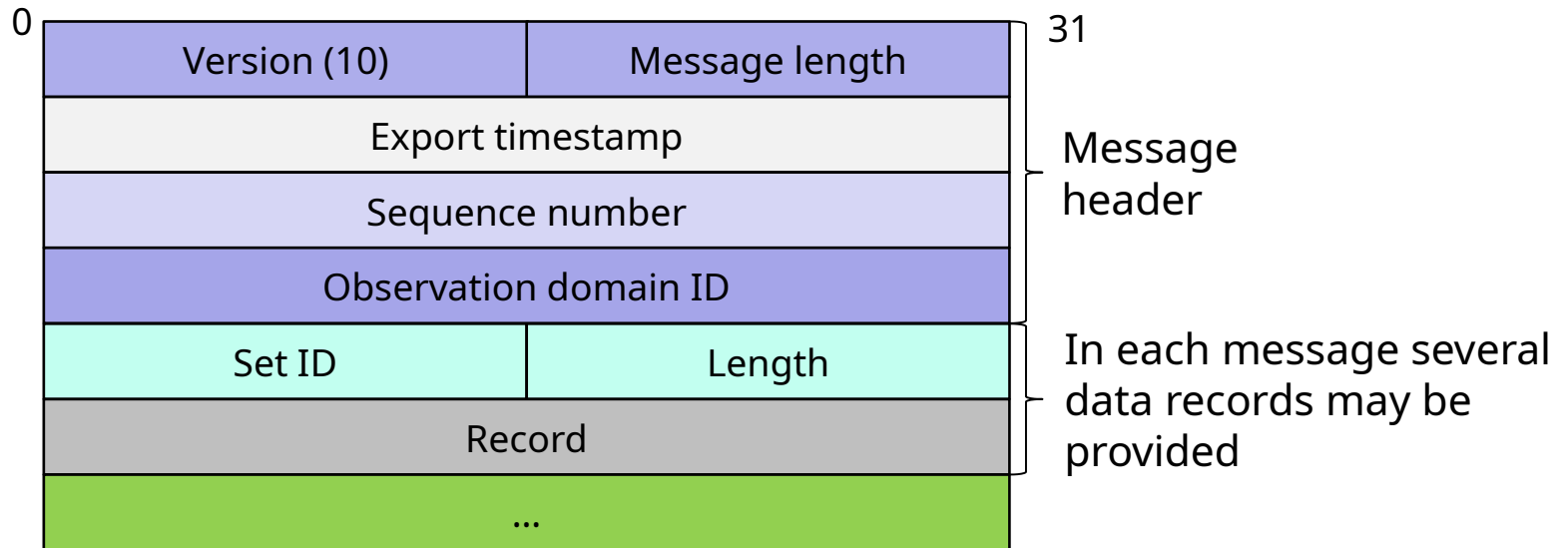
Internet Protocol Flow Information Export (IPFIX)



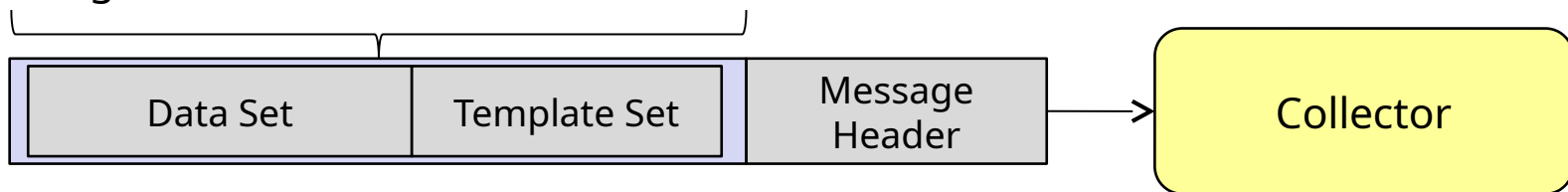
- IPFIX covers a push protocol specified in RFCs 7011 – 7015 for transmitting data flows to a central entity
- Specified based on NetFlow
- Definition of a "Flow" in RFC 7011: "A Flow is defined as a set of packets or frames passing an Observation Point in the network during a certain time interval. All packets belonging to a particular Flow have a set of common properties."
- Instances of a Flow are named Flow Records, e.g. amount of IP packets exchanged between two specific UDP endpoints



IPFIX – Protocol details



Set consists of a set ID (e.g. 2 for template set), length information and entries



- Templates describing the structure of the data may be transferred together with this data
- By using templates, the sender can transmit information for interpreting data of unknown internal structure → increases flexibility of approach

IPFIX - Example

- Example shows a template record and a data record
- Data record specifies the amount of IP packets transmitted from a specific IP to a specific destination IP address

Set ID = 2	Length = 20	
Template ID = 260	Amount of fields = 3	
Type = destinationIPv4Address(12)	Length = 4 Byte	• Type defined by IANA or "private" (=context specific types) may be used
Type = sourceIPv4Address (8)	Length = 4 Byte	
Type = packetDeltaCount (3)	Length = 8 Byte	
Set ID = 260	Length = 20	
Record 1, field 1 = 141.76.40.30.1		• Data record provided with reference to template with ID 260
Record 1, field 2 = 8.8.8.8		
Record 1, field 3 = 42 Pakete		

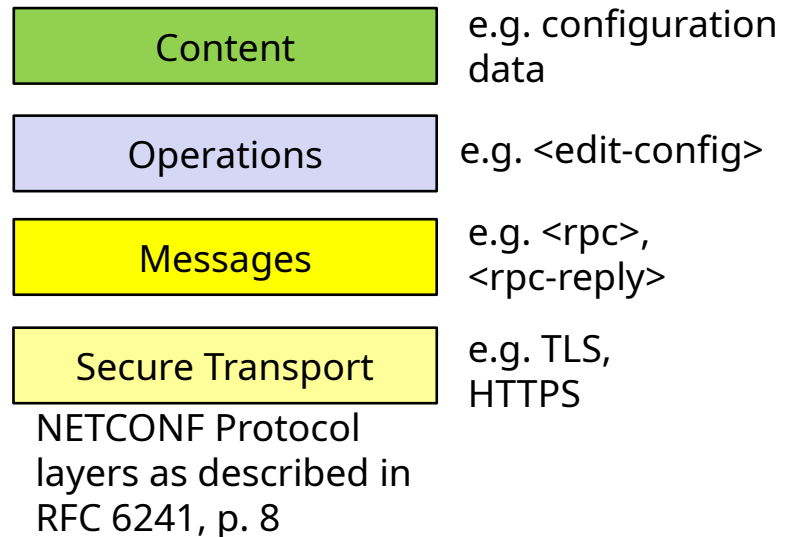
- For wide-spread types used within template records, IANA specified relevant values, see:
 - <http://www.iana.org/assignments/ipfix/ipfix.xhtml>
- Several data records referring to the same template may be provided

NETCONF

- SNMP is very wide-spread, however, it has many essential drawbacks
 - For an overview of such drawbacks see RFC 3535, e.g.:
 - *"The SNMP transactional model and the protocol constraints make it more complex to implement MIBs, as compared to the implementation of commands of a command line interface interpreter."*
 - *"There is too little deployment of writable MIB modules."*
 - *"SNMP does not support easy retrieval and playback of configurations."*
- **For configuring network devices, CLIs are dominant**

- Goals of **NETCONF**:

- Provide a protocol for consistently initiate, modify and delete network configurations



NETCONF

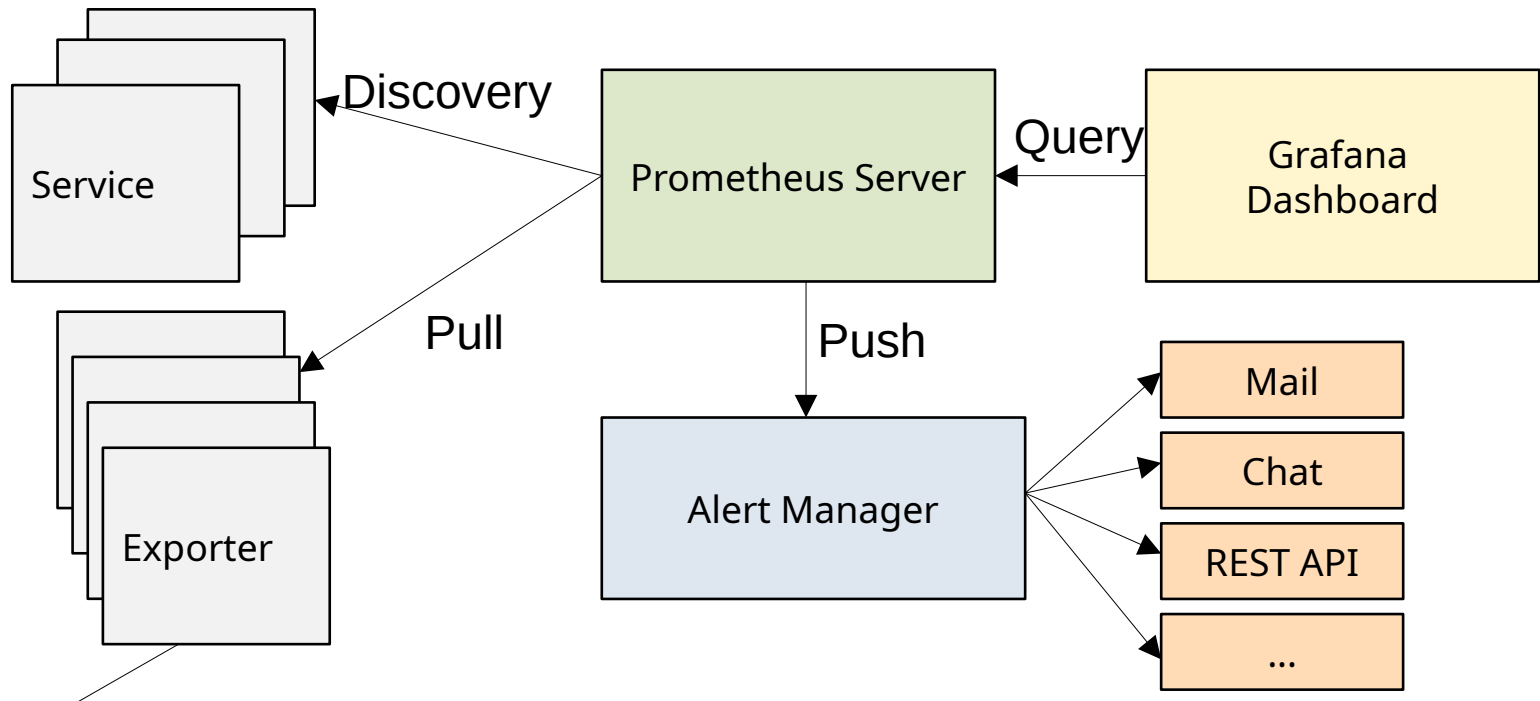
- Configuration is done based on an RPC mechanism for exchanging XML data
- Base protocol offers nine operations (see section 7 in RFC 6241):
 - get, get-config, edit-config, copy-config, delete-config, lock, unlock, close-session, kill-session
- Differentiation between configuration and state data
- Protocol provides possibility to lock *Configuration Datastore*

Example
taken from
RFC 6241:
Set the MTU
of an
Ethernet
interfaces to
1500

```
<rpc message-id="101" xmlns="urn:ietf:params:xml:ns:netconf:base:1.0"> <edit-  
config> ...  
<config>  
  <top xmlns="http://example.com/schema/1.2/config">  
    <interface>  
      <name>Ethernet0/0</name>  
      <mtu>1500</mtu>  
    </interface>  
  </top>  
</config>  
</edit-config>  
</rpc>
```

Prometheus

- Prometheus is a often used fundament for monitoring and alerting systems
- Core of Prometheus is a time-series database for storing data acquired from observed entities
- Available under Apache 2.0 license



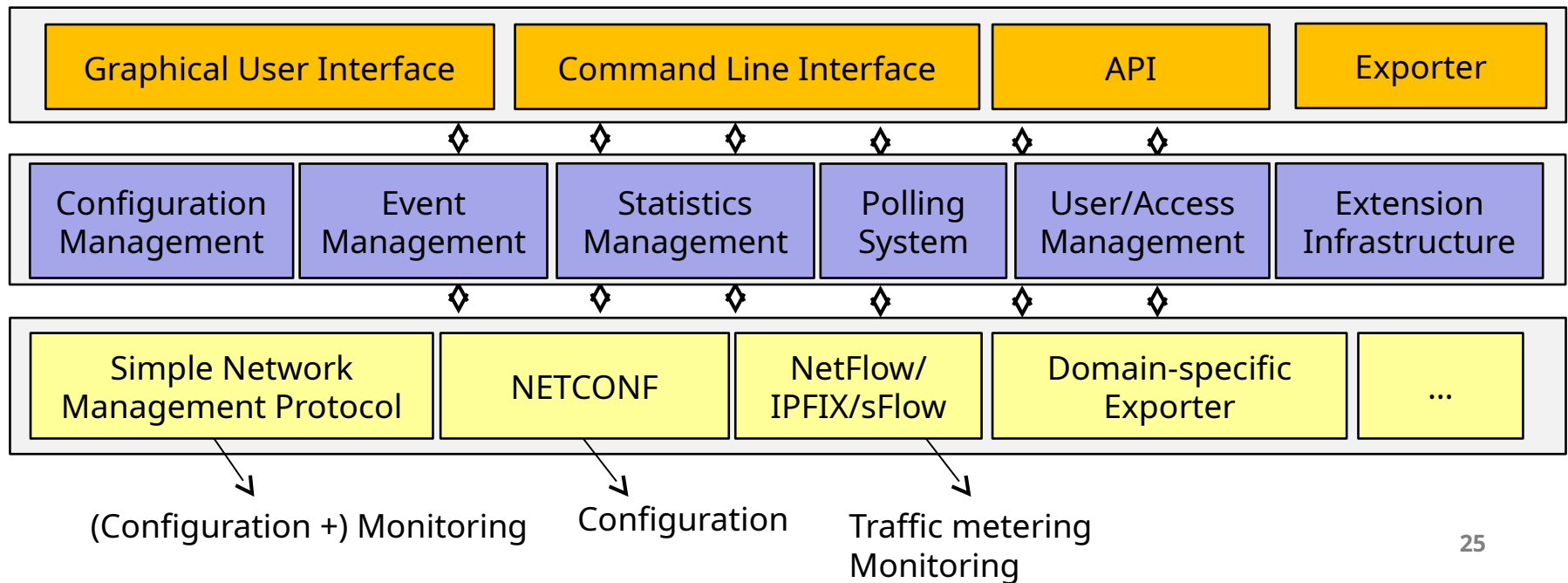
E.g. based on regular system utilities or SNMP

Summary

- F** Fault Management
- C** Configuration Management
- A** Accounting Management
- P** Performance Management
- S** Security Management

Management protocol specifications may cover:

- Schema description language
- Data schema
- Communication protocol
- Security mechanisms and protocols



References

Selected RFCs

- Structure and Identification of Management Information for TCP/IP-based Internets <https://tools.ietf.org/html/rfc1155>
- Structure of Management Information Version 2 <https://tools.ietf.org/html/rfc2578>
- Specification of the IP Flow Information Export (IPFIX) Protocol <https://tools.ietf.org/html/rfc7011>
- Overview of the 2002 IAB Network Management Workshop <https://tools.ietf.org/html/rfc3535>
- Network Configuration Protocol (NETCONF) <https://tools.ietf.org/html/rfc6241>